

Excellent!

$\frac{9}{9} \Rightarrow 100\%$

0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	10
---	-----	---	-----	---	-----	---	-----	---	-----	---	-----	---	-----	---	-----	---	-----	---	----

Name: _____

These questions assess your understanding of the material from Chapters 22 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. The hot and neutral wires supplying DC power to a light-rail commuter train carry 890 A and are separated by 66 cm. What is the magnitude of the force between 80 m of these wires?

$$\frac{F}{l} = \frac{\mu_0 I^2}{2\pi r} \Rightarrow F = l \frac{\mu_0 I^2}{2\pi r}$$

ANSWER: 19.202 N ✓

$$F = 80 \frac{(4\pi \times 10^{-7})(890)^2}{2\pi(0.66)} = 19.202 \text{ N}$$

2. A wire carries a 18 A current toward the south. If a large magnet were placed directly to the west of the wire and emits a 1.0 T magnetic field perpendicularly towards the wire, calculate the force exerted on the wire per length of wire.

$$F = I l B \sin(90)$$

ANSWER: 18 $\frac{\text{N}}{\text{m}}$ ✓

$$\Rightarrow \frac{F}{l} = IB = (18)(1.0) \Rightarrow \frac{F}{l} = 18 \frac{\text{N}}{\text{m}}$$

3. A 0.12-kg baseball, pitched at 33 $\frac{\text{m}}{\text{s}}$ horizontally and perpendicularly to Earth's horizontal 35- μT field, has a 14-nC charge on it. What distance is the ball deflected from its path by the magnetic force after it travels 8.1 m horizontally? Do not consider the displacement due to the weight force. You may assume that the ball does not hit the ground during this motion.

$$v = \frac{\Delta x}{\Delta t} \Rightarrow \Delta t = \frac{\Delta x}{v} = \frac{8.1}{33} = 0.24545... \text{ s} \quad \text{ANSWER: } \underline{4.059 \times 10^{-12} \text{ m}} \quad \checkmark$$

$$F = qvB \sin 90 \quad F = (35 \times 10^{-6})(33)(14 \times 10^{-9})$$

$$= 1.617 \times 10^{-11} \text{ N} \quad F = ma \Rightarrow F/l = a$$

$$\frac{1.617 \times 10^{-11}}{0.12} = 1.3475 \times 10^{-10}$$

$$y_f = v_0 t + \frac{1}{2} a t^2 \Rightarrow y_f = \frac{1}{2} a t^2$$

$$y_f = \frac{1}{2} (1.3475 \times 10^{-10})(0.24545)^2 = 4.059 \times 10^{-12}$$

+3

4. A Hall-effect flow probe is placed on an artery, applying a 0.076 T magnetic field across it. What is the Hall emf, given the vessel's inside diameter is 6.1 mm and the average velocity of the blood is 73 $\frac{\text{cm}}{\text{s}}$?

$$\mathcal{E} = B l v$$

$$\mathcal{E} = (0.076)(6.1 \times 10^{-3})(0.73) = 3.38428 \times 10^{-4} \text{ V}$$

ANSWER: $3.38 \times 10^{-4} \text{ V}$

$$\frac{73 \text{ cm}}{\text{s}} \left| \frac{\text{m}}{100 \text{ cm}} \right| = 0.73 \text{ m/s}$$

5. Find the maximum torque on a 190-turn square loop of wire that is 3.3 cm on each side that carries 11. A of current in a 0.45-T field.

$$\tau = N I A B \sin 90$$

$$\tau = (190)(11)(0.033^2)(0.45) = 1.024 \text{ N}\cdot\text{m}$$

ANSWER: $1.024 \text{ N}\cdot\text{m}$

6. A proton travels with speed $1.4 \times 10^6 \frac{\text{m}}{\text{s}}$ at an angle of 19° to a uniform magnetic field. If its path has a circular component of radius 0.059 m, calculate the magnitude of the magnetic field.

$$r = \frac{m v \sin(\theta)}{q B} \Rightarrow B = \frac{m v \sin(\theta)}{q r}$$

$$B = \frac{(1.67 \times 10^{-27})(1.4 \times 10^6)(\sin 19)}{(1.6 \times 10^{-19})(0.059)} = 0.0806 \text{ T}$$

ANSWER: 0.0806 T

7. Find the current that must run through a current-carrying circular loop so that it produces a 70.- μT magnetic field at its center, given that the radius of the circular loop is 20. cm.

$$B = \frac{\mu_0 I}{2 R} \Rightarrow \frac{B 2 R}{\mu_0} = I$$

$$I = \frac{(70 \times 10^{-6})(2)(0.20)}{(4\pi \times 10^{-7})} = 22.28 \text{ A}$$

ANSWER: 22.28 A

74

8. A large water main is 1.3 m in diameter and the average water velocity is $0.46 \frac{\text{m}}{\text{s}}$. Find the Hall voltage produced if the pipe runs perpendicular to Earth's 4.5×10^{-5} -T field.

$$E = B L v = (4.5 \times 10^{-5})(1.3)(0.46) \quad \text{ANSWER: } \underline{2.691 \times 10^{-5} \text{ V}}$$
$$E = 2.691 \times 10^{-5} \text{ V}$$

9. Calculate the radius of curvature of the path of a proton having a velocity of $1.5 \times 10^7 \frac{\text{m}}{\text{s}}$ perpendicular to a magnetic field of strength $B = 0.69 \text{ T}$.

$$r = \frac{mv}{qB} = \frac{(1.67 \times 10^{-27})(1.5 \times 10^7)}{(1.60 \times 10^{-19})(0.69)} \quad \text{ANSWER: } \underline{0.2269 \text{ m}}$$
$$= 0.2269 \text{ m}$$

+2

1. $F = 19.2 \text{ N} \therefore \frac{F}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi r}$
2. $\frac{F}{\ell} = 18.0 \frac{\text{N}}{\text{m}} \therefore F = I\ell B \sin \theta, \theta = 90^\circ$
3. $\Delta y = 4.06 \times 10^{-12} \text{ m} \therefore F = qvb \sin \theta, F_{\text{net}} = ma, \vec{x}(t) = \vec{x}_0 + \vec{v}_0 t + \frac{1}{2} \vec{a} t^2$
4. $\varepsilon = 3.38 \times 10^{-4} \text{ V} \therefore \varepsilon = B\ell v$
5. $\tau = 1.02 \text{ N} \cdot \text{m} \therefore \tau = NIAB \sin \theta \therefore \tau_{\text{max}} = NIAB$
6. $B = 0.0806 \text{ T} \therefore r = \frac{mv}{qB}, m_p = 1.67 \times 10^{-27} \text{ kg}, q_p = 1.60 \times 10^{-19} \text{ C}$
7. $I = 22.3 \text{ A} \therefore B = \frac{\mu_0 I}{2R}$
8. $\varepsilon = 2.69 \times 10^{-5} \text{ V} \therefore \varepsilon = B\ell v$
9. $r = 0.227 \text{ m} \therefore r = \frac{mv}{qB}, m_p = 1.67 \times 10^{-27} \text{ kg}, q_p = 1.60 \times 10^{-19} \text{ C}$